# Dennis G. Pezan

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## Education

**United States Military Academy at West Point** — West Point, NY
*B.S., Operations Research (Finance Specialization), Minor in Artificial Intelligence* — Expected May 2029
- CQPA: 3.571 | Dean's List, Fall 2025 and Spring 2026
- Research Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply (Advisor: LTC Matthew Webb)

## Research Experience

**Contested Logistics / Network Interdiction Research – D/Math, USMA** — West Point, NY
*Undergraduate Researcher (Advisor: LTC Matthew Webb)* — Jul 2026 – Present
- Extend LTC Webb's doctoral research on reinforcement learning for maximum-flow network interdiction to a 57-node, 79-edge open-source model of Taiwan's crude oil import network with sourced capacities and seven costed adversary actions anchored on historical events
- Connected the network to a reinforcement-learning interdiction environment, verified it against an independent max-flow calculation, and fixed a targeting-filter issue that could hide the Strait of Hormuz
- Score each adversary plan by the expected share of Taiwan's crude demand still delivered after rerouting; the agent must beat random plans (about 90%) and a greedy baseline (70%) and approach an exact solver across six sensitivity cases

**Reinforcement Learning for Contested Resupply – TDAC, U.S. Army** — Fort Lee, VA
*Research Intern, Modeling & Analysis Division (AIAD; Mentor: MAJ Erica Mikelson)* — Jul 2026
- Redesigned the reward function for a proximal policy optimization (PPO) reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's stochastic sustainment simulation; trained and evaluated the agent in Python (PyTorch)
- Built two diagnostic tools that exposed four reward flaws, including an agent that outscored every baseline while starving the priority unit; the retrained agent more than tripled the share of requests filled and beat a hand-coded priority-aware baseline across 101 random seeds
- Presented findings to Branch Chief LTC Ryan Miller at Fort Lee and, by call, to TDAC's AI and reinforcement-learning researchers (LTC Joseph Pedersen) at the Naval Postgraduate School in Monterey, CA

**Tree-Based Methods in Machine Learning – D/Math, USMA** — West Point, NY
*Independent Study in Machine Learning (MA289), Advisor: LTC Patrick Kuiper* — Oct 2025 – May 2026
- Designed and completed an independent study of decision trees, bagging, and random forests, and presented the findings to officers
- Built an interactive random forest research site with a live classifier demonstration and hyperparameter explorer

## Projects

**Tree-Based Methods Research Site** – dennis-pezan.com/ma289.html — 2026
- Trained and evaluated decision trees, bagging, and random forests on the Titanic dataset in Python (scikit-learn) on an identical 80/20 split
- Computed pairwise tree correlation, overfitting gaps, and feature importance to show how ensemble decorrelation reduces variance; deployed via GitHub and Vercel

**CFT Packing List** – dennis-pezan.com/cft.html — 2026
- JavaScript web tool that helps cadets organize and track gear for Cadet Field Training; saves progress in the browser and deploys on Vercel from GitHub

**Reward Engineering Case Study** – dennis-pezan.com/rl-reward.html — 2026
- Public write-up diagnosing and correcting reward hacking in a simulated shipping network and raising requests met from 24% to 81%

**Mounger Writing Center Website Redesign** – dennis-pezan.com/mwc-redesign.html — Aug 2026 – Present
- Independent redesign concept for West Point's writing center site, built in Next.js, TypeScript, and Tailwind CSS with a custom WebGL ink shader and editable content files for staff

## Professional Development

**Technology and Innovation AIAD – Department of Social Sciences, USMA** — California and Colorado
*Summer AIAD on defense technology (separate from the academic-year Sea to Shining Sea Fellowship)* — Jun 2026
- Held in-depth discussions with leaders at Palantir, Stanford's Gordian Knot Center, OpenAI, Apple, and Meta Reality Labs on how defense technology is developed, procured, and fielded
- Topics: startup barriers to Army adoption, agentic AI and large language models for defense, supply-chain risk, and AI regulation
- Observed how the 4th Infantry Division employs and gives feedback on these systems under Next Generation Command and Control; visited U.S. Northern Command, NORAD, and the Cheyenne Mountain Complex

## Honors & Awards

**Dean's List**, United States Military Academy — Fall 2025, Spring 2026
**Stokes Writing Fellowship**, United States Military Academy — 2026–Present
- Endowed fellowship funding research travel, expert mentorship, and publication of original research

**Norwegian Foot March Badge**, Norwegian Armed Forces — 2026
**Sea to Shining Sea Fellowship**, United States Military Academy — 2026–Present
- Academic-year fellowship engaging cadets with senior leaders in national security, government, finance, and technology; separate from the summer Technology and Innovation AIAD

## Leadership & Service

**Center for Enhanced Performance, USMA** — West Point, NY
*Cadet-in-Charge, MA103 Data-Driven Modeling Tutoring* — Aug 2026 – Present
- Lead the full tutor staff for MA103, a core mathematics course with 1,010 cadets enrolled; direct tutoring coverage for every major graded event
- Built a study site (dennis-pezan.com/ma103-tutoring.html) with two full-length practice exams, a mirror of a graded gate, and a 59-drill reference card, each with an answer key

**Mounger Writing Center, USMA** — West Point, NY
*Writing Consultant* — Aug 2026 – Present
- Selected as a Stokes Writing Fellow to coach cadets on strengthening the rhetoric and argument of their writing

**Center for Leader Development in STEM (CLDSTEM), USMA** — West Point, NY
*Member* — Oct 2025 – Present
- Bring hands-on STEM projects to students, reaching 800+ students across the Chicagoland and Greater Los Angeles areas
- Selected to facilitate leadership and ethical decision-making workshops and a Q&A panel for 150+ high school students at a West Point Leaders of Character Seminar in Mesquite, TX, with the Directorate of Admissions (Nov 2026)

**Cadet Ambassador Program, USMA** — West Point, NY
*Cadet Ambassador* — Jan 2026 – Present
- Represent West Point in outreach to prospective cadets and help foster a welcoming environment for future applicants

**Finance Club, USMA** — West Point, NY
*Member* — Oct 2025 – Present
- Engage with industry executives from companies including Palantir, Uber, and Meta on leadership in business, business ethics, and personal finance

**Recycle & Reboot** — Plainfield, IL
*Volunteer Senior Publicist* — May 2024 – Aug 2024
- Led outreach to secure donated technology; helped refurbish and distribute 100+ computers to underserved communities

**Illinois Premier Boys State** — Illinois
*Candidate* — Jun 2024
- Elected Alderman and Township Supervisor; advanced to the Lieutenant Governor primary through competitive public speaking

## Professional Experience

**McDonald's** — Plainfield, IL
*Department Manager* — Sep 2022 – Jul 2024
- Promoted at age 16 as the region's youngest department manager; sustained 90%+ customer satisfaction scores
- Scheduled 80+ employees weekly with 95% on-time accuracy; oversaw $20,000+ in weekly supply procurement

**GrayStone AI Solutions** — Remote
*Intern* — Jun 2024 – Jul 2024
- Built Python and Excel tools to support financial planning and analysis workflows

## Skills

**Technical:** Python (PyTorch, scikit-learn), SQL, Java, JavaScript/TypeScript (Next.js) | **Methods:** reinforcement learning, network optimization, machine learning
**Certifications:** JPMorgan Chase Software Engineering Job Simulation (2026); Microsoft Fabric Data Architecture (2026)
**Languages:** Spanish (fluent) | **Clearance:** Secret